Systems Analysis Final Project Template: (Updated 12/1/2025)

*Note: Please cross-reference with the **Volere Requirements Template***

Project Name: E.A.R.T.H

Author: Jake Tomesh **Date**: 12/3/2025

Requirements Document Details (Parts 1-10 are from the Volere Template Specification)

1: Purpose of the Work:

The purpose of the E.A.R.T.H (Environmental Analytics & Real-Time Hub) system is to provide a centralized platform for businesses to track, validate, convert, store, display, and forecast greenhouse gas emission data. The system standardizes emissions from multiple sources into CO_2e baseline units, enables threshold-based alerting, and supports informed sustainability decision-making through visual dashboards and predictive modeling.

2: List of Stakeholders:

Andy Bangsberg, Jeff Rybacki, Systems Administrators, Data Analysts, Sustainability Leadership

3: Constraints

3a-3g if applicable
Must be completed by: 5/11/2026
Must use at least one API
Must use MVC
Must use at least five tables
Must do some data logging
Must use encryption for storing passwords (Salt and Hash)
Website must be responsive
Must comply with GHG Protocol standards.
Must support at least 100 concurrent users when entering emissions data.
Must support screen sizes as small as an IPad Mini.

4**: Naming Conventions and Terminology** (4.a)

- **CO_2e:** Carbon dioxide equivalent; baseline unit for emissions comparison.
- **Emission Type:** The category of emission source (e.g., electricity, fuel).
- **Emission Factor:** The number by which physical emissions are multiplied by to convert to CO_2e
- **Baseline Unit Conversion:** Process of converting all emissions into CO_2e.
- **Threshold Limit:** Admin-configured CO_2e value defining upper limits.
- **Dashboard:** Primary interface for general navigation.

5: Relevant Facts and Assumptions

5a. Relevant Facts

- The business tracks multiple emission types in different units.
- All emissions must be converted to a consistent baseline (CO_2e).
- Thresholds can differ by emission category.

- Users need visual dashboards for interpretation.

5b. Business Rules

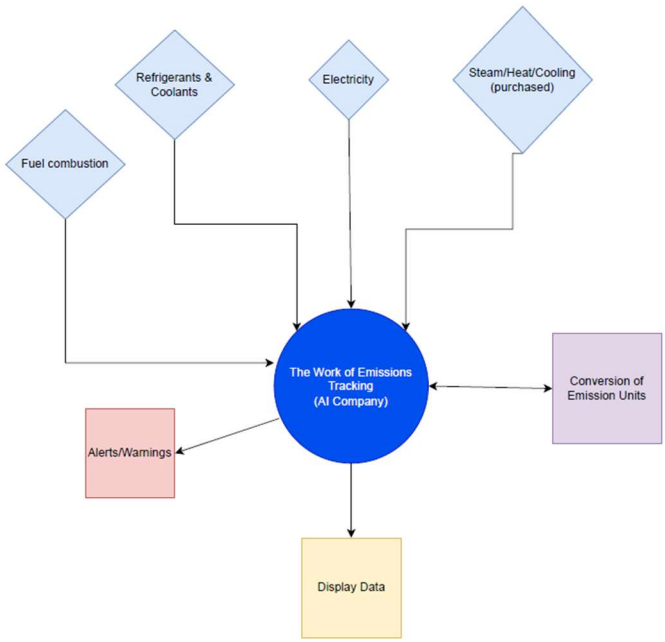
- All emissions data must be validated before acceptance.
- Data must convert to CO_2e with two-decimal accuracy.
- Threshold breaches must trigger pop-up alerts.
- All stored data must include timestamps and identifiers.
- Dashboards must highlight threshold and alert logs.

5c. Assumptions

- Emission factors used for conversions are accurate and updated as needed.
- Users entering data understand the emission categories they're selecting.
- Internet connectivity is stable enough for real-time dashboard updates.
- This product is deployed per business as a standalone product.

6: Scope of Work

6b. Context Diagram



6c. Business Event List

Business Event List

# Event Name	**Input** (if not obvious) **Output** (if not obvious)
User enters emissions data	**I:**Emission quantity & unit. **O:**Validated or rejected.
System converts data to CO_2e	**I:**Validated raw emissions **O:**Standardized CO_2e value

System tests converted CO_2e against thresholds	**I:**Converted CO_2e **O:**Alert flag & notifications
System stores emissions data	**I:**Valid CO_2e record **O:**Database entry insert/update
System displays data	**I:**Stored & forecasted values **O:**Visual tables & graphs
System facilitates a user login	**I:**User email & password **O:**User is validated/invalidated
System facilitates user registration	**I:**Admin registers new user information **O:**User data exists in database for login
System grants admin access to set alert thresholds	**I:**Threshold unit & amount **O:**Threshold stored in database & actively monitored

7a. Business Data Model (list of objects you will need. Example: Customer, Order, Product)

a. EarthUser
b. BaseUnitType
c. EmissionLog
d. EmissionType
e. EmissionFactor
f. UnitType
g. ThresholdLimit
h. AlertLog
i. RoleType
j. Licensee

7b: Data Dictionary (used to help define any data that is important or not obvious)

a. User - Represents a person who can log into the system (admin, analyst, etc.).
b. EmissionLog – Contains all data related to a single emissions entry.
c. EmissionType – Provides the valid types of emissions to be recorded.
d. UnitType – Provides the valid types of measurable units to be recorded.
e. EmissionFactor – Contains the "math" necessary to convert units into CO_2e.
f. ThresholdLimit – Represents configurable CO_2e limits for alerts.
g. AlertLog – Contains all data related to a triggered alert.
h. Licensee – Contains IDs for multiple companies to use the product.

8: List of User Stories with step1, step2, step3….. (this should cover 90%+ of the requirements)

Story 1 — Enter Emissions Data

- As a: Data Entry User
- I can: Enter emissions quantity, type, and units into the system.
- Rationale: To record organizational emissions across all tracked sources.
- Fit Criterion: Input is validated and either accepted or rejected with an error.

Story 2 — Convert Emissions to CO_2e

- As a: System
- I can: Convert all validated emissions into baseline CO_2e units.
- Rationale: Standardization of all data for comparison and aggregation.
- Fit Criterion: All conversions stored to two decimal places.

Story 3 — Trigger Alerts

- As an: Administrator
- I can: Receive dashboard and email alerts when thresholds are exceeded.
- Rationale: To take corrective actions promptly.
- Fit Criterion: Alerts appear whenever CO_2e surpasses limits.

Story 4 — Store Emissions

- As a: Database system
- I can: Save validated + converted emissions with timestamps.
- Rationale: For historical reporting and auditing.
- Fit Criterion: Data persists and can be queried later.

Story 5 — Display Dashboard

- As a: Data Analyst
- I can: See tables, graphs, and threshold statuses.
- Rationale: To interpret emissions trends visually.
- Fit Criterion: Color-coded indicators for thresholds.

Story 6 — User Role

- As a: User
- I can: Login to the application.
- Rationale: Authentication is enacted to ensure the safety of data.
- Fit Criterion: User is validated and granted access to the application.

Story 7 — User Role

- As a: User
- I can: Make emission entries and view data models.
- Rationale: To facilitate the flow of data and analyze output.
- Fit Criterion: User restrictions to all tools beyond the scope of the user role.

Story 8 — Admin Role

- As an: Admin
- I can: Register a user.
- Rationale: To add user data to the database for new user authentication.
- Fit Criterion: New users are added to the database and able to be authenticated.

Story 9 — Admin Role

- As an: Admin
- I can: Make emissions entries, view data models, CRUD threshold data.

- Rationale: To give permission to specific users entrusted within the company to make changes to the application logic.
- Fit Criterion: Admins set thresholds in addition to all user abilities.

Story 10 — Licensee

- As an: Admin
- I can: Assign a company license to a user
- Rationale: To ensure that users are affiliated with the correct company and its data.
- Fit Criterion: Users will only have access to their companies data.

9 & 10:

For any additional requirements (Functional or Non-Functional) use the abbreviated Snow card format for the requirements. **(Note: You will also add these as cards in Trello)**
You can also simply write user stories.

9: Any additional **Functional** requirements not identified in the list of user stories.

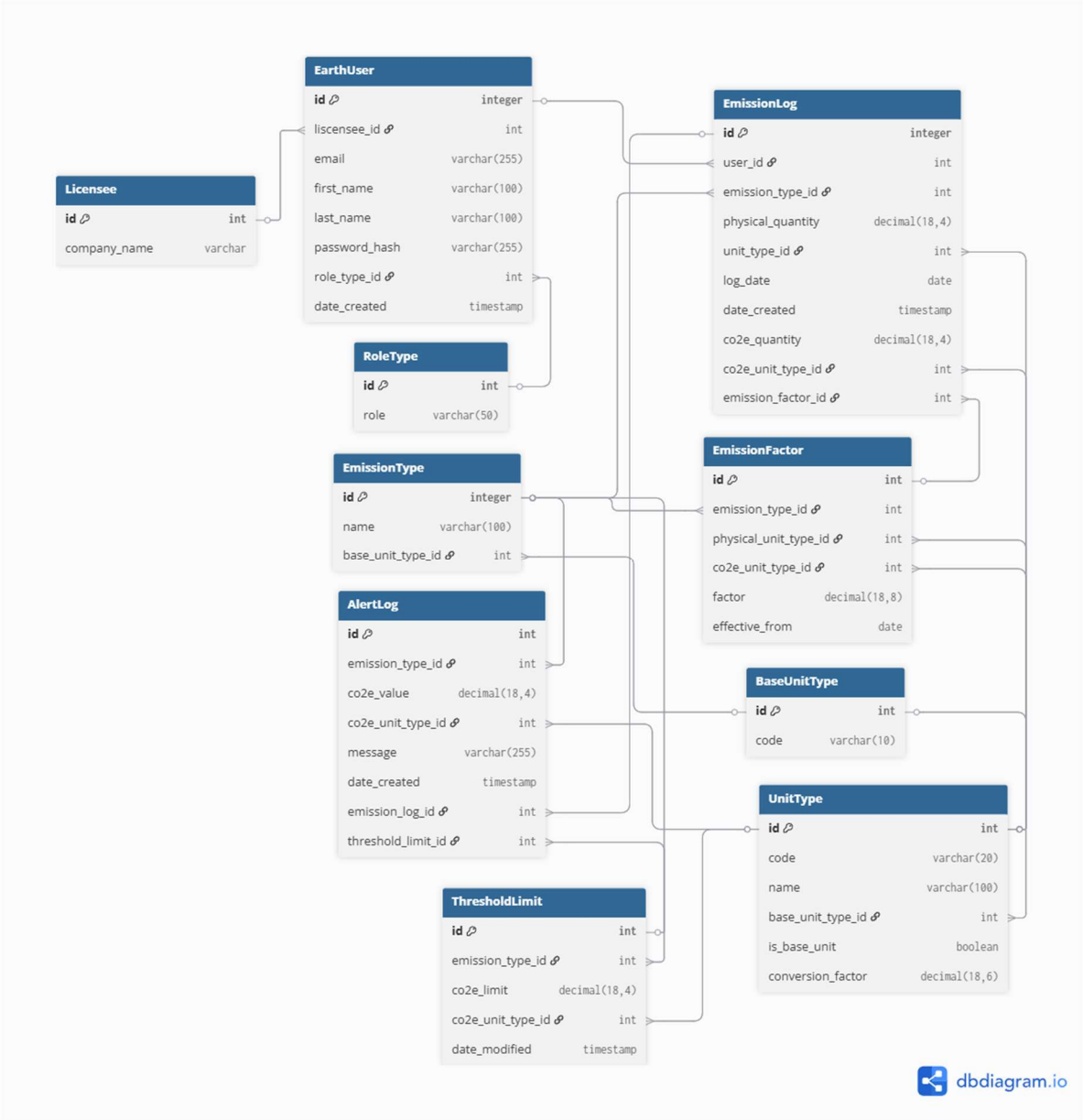
a. Admins can modify emission factors
   As a: Administrator
   I can: Update and change emission factors
   Rationale: To stay current with real-world implementation of emission factors
   Fit Criterion: Emission conversion calculations result in accurate outputs via the emission factor being used.

10: Any additional **Non-functional** requirements not identified in the list of user stories.

a. System shall timeout sessions after 15 minutes of inactivity.
b. System shall generate emissions reports within 10 seconds (for up to one years worth of data).

Appendix A:

Copy of Database Diagram (Normally, this is not a requirement, but I want a copy to see how it will compare to your finished product. Also, it can be helpful to remind you of the fields you might need in the product).



Appendix B:

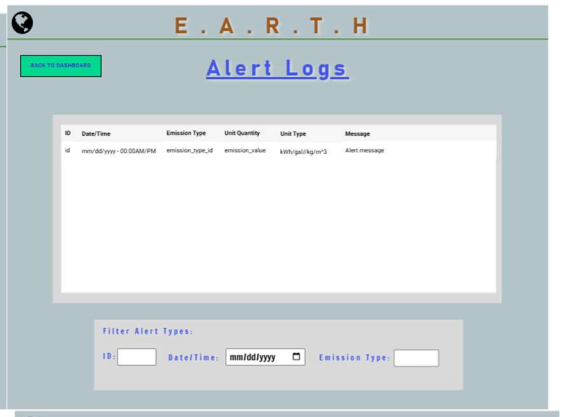
URL to Trello Board

https://trello.com/b/DP5YNTw5/tomesh-ourappsdevearth

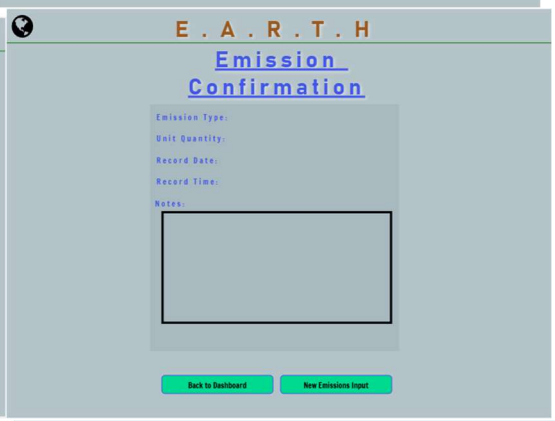
Appendix C:

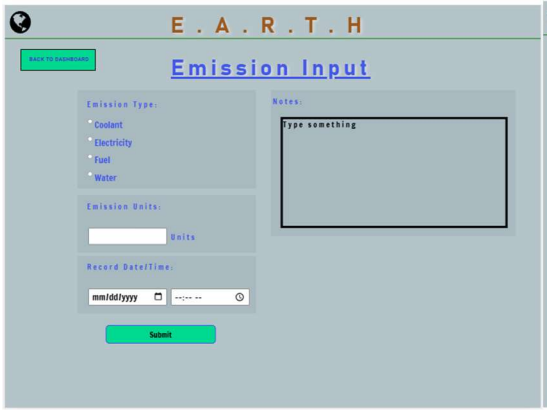
Include all screenshots of your PDF of your Prototype

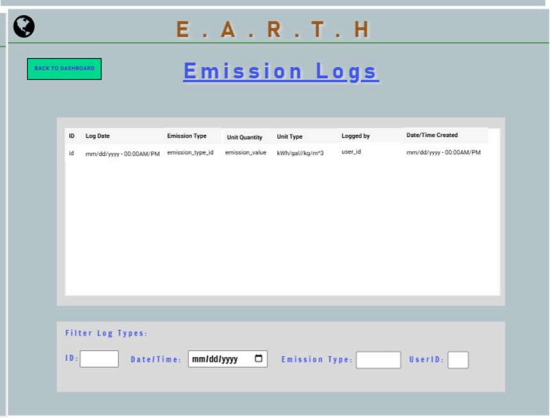
E . A . R . T . H
BACK TO DASHBOARD
Register User
User Information:
First Name:
Last Name:
Email:
Password:
Role Type:
Submit

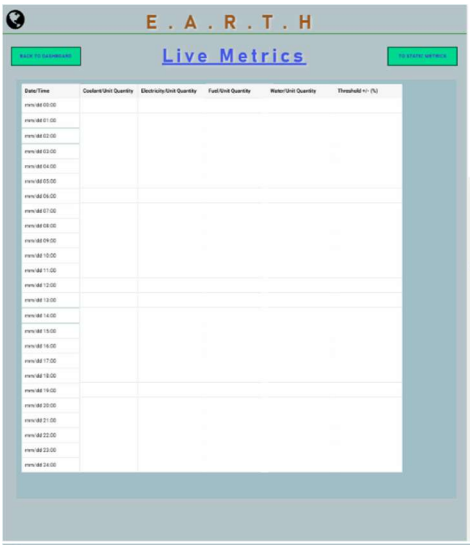

E . A . R . T . H
BACK TO DASHBOARD
Alert Logs
ID
Date/Time
Emission Type
Unit Quantity
Unit Type
Message
id
mm/dd/yyyy - 00:00AM/PM
emission_type_id
emission_value
kWh/gal/kg/m^3
Alert message
Filter Alert Types:
ID:
Date/Time:
mm/dd/yyyy
Emission Type:

E . A . R . T . H
Help us save E.A.R.T.H,
Emissions Input
View Metrics
Alert Logs
Emissions Logs
Admin Portal


E . A . R . T . H
Emission Confirmation
Emission Type:
Unit Quantity:
Record Date:
Record Time:
Notes:
Back to Dashboard
New Emissions Input


E . A . R . T . H
BACK TO DASHBOARD
Emission Input
Emission Type:
Coolant
Electricity
Fuel
Water
Emission Units:
Units
Record Date/Time:
mm/dd/yyyy
--:-- --
Notes:
Type something
Submit


E . A . R . T . H
BACK TO DASHBOARD
Emission Logs
ID
Log Date
Emission Type
Unit Quantity
Unit Type
Logged by
Date/Time Created
id
mm/dd/yyyy - 00:00AM/PM
emission_type_id
emission_value
kWh/gal/kg/m^3
user_id
mm/dd/yyyy - 00:00AM/PM
Filter Log Types:
ID:
Date/Time:
mm/dd/yyyy
Emission Type:
UserID:


E . A . R . T . H
BACK TO DASHBOARD
Live Metrics
TO STATIC METRICS
Date/Time
Coolant/Unit Quantity
Electricity/Unit Quantity
Fuel/Unit Quantity
Water/Unit Quantity
Threshold +/- (%)
mm/dd 00:00
mm/dd 01:00
mm/dd 02:00
mm/dd 03:00
mm/dd 04:00
mm/dd 05:00
mm/dd 06:00
mm/dd 07:00
mm/dd 08:00
mm/dd 09:00
mm/dd 10:00
mm/dd 11:00
mm/dd 12:00
mm/dd 13:00
mm/dd 14:00
mm/dd 15:00
mm/dd 16:00
mm/dd 17:00
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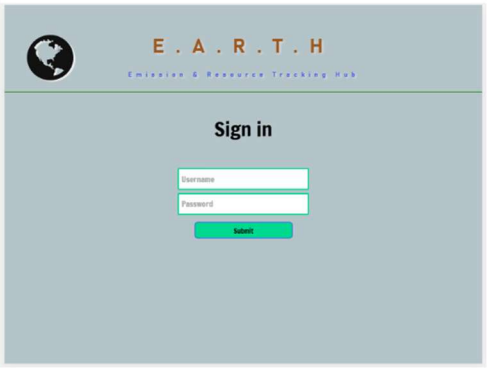
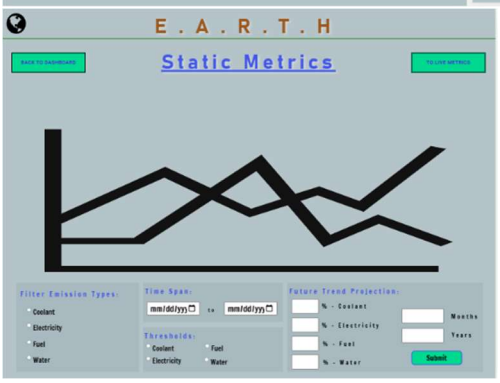
mm/dd 24:00


E . A . R . T . H
Emission & Resource Tracking Hub
Sign in
Username
Password
Submit


E . A . R . T . H
BACK TO DASHBOARD
Static Metrics
TO LIVE METRICS
Filter Emission Types:
Coolant
Electricity
Fuel
Water
Time Span:
mm/dd/yyyy
to
mm/dd/yyyy
Thresholds:
Coolant
Fuel
Electricity
Water
Future Trend Projection:
% - Coolant
% - Electricity
% - Fuel
% - Water
Months
Years
Submit